Reinforcement Learning

from one delivery van to the Bellman equation

Carlos Cotrini, ETH Zurich

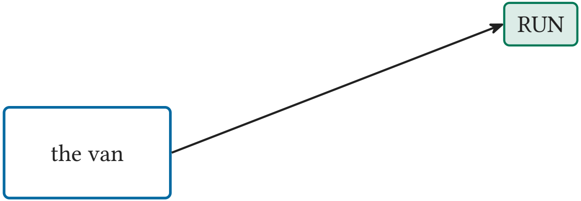
Motivation

One van, one call a week

the van

every week: exactly one call

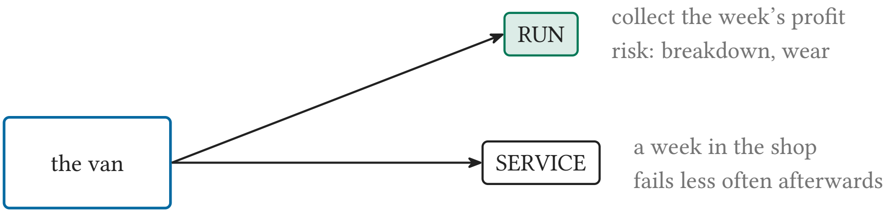
One van, one call a week



collect the week's profit
risk: breakdown, wear

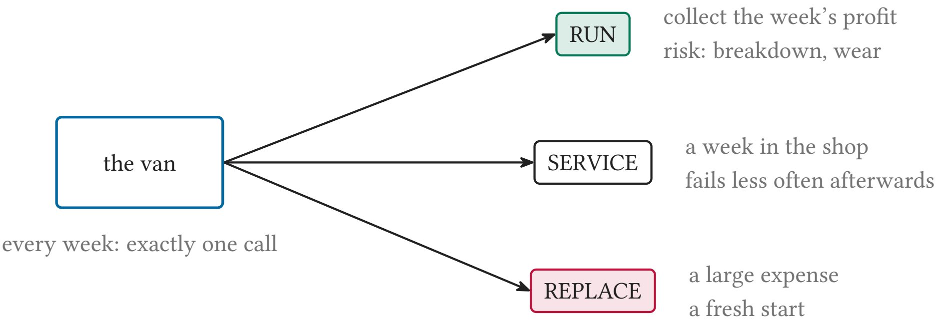
every week: exactly one call

One van, one call a week

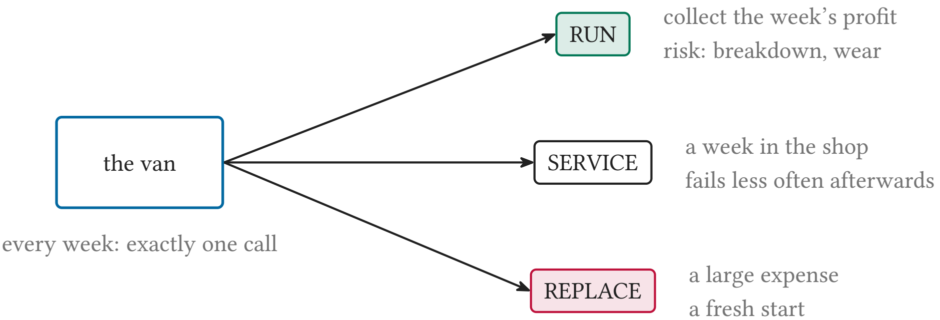


every week: exactly one call

One van, one call a week



One van, one call a week



short-term cost against long-term payoff, chance at every step

Markov decision processes

The model: four ingredients

S: states

four wear
levels

The model: four ingredients

S: states

four wear
levels

A: actions

three
calls

The model: four ingredients

S: states	A: actions	T: dynamics
four wear levels	three calls	how the world reacts

The model: four ingredients

S: states	A: actions	T: dynamics	$\gamma = 0.9$
four wear levels	three calls	how the world reacts	how much the future counts

The model: four ingredients

S: states	A: actions	T: dynamics	$\gamma = 0.9$
four wear levels	three calls	how the world reacts	how much the future counts

$$(s', r) = T(s, a)$$

The model: four ingredients

S: states	A: actions	T: dynamics	$\gamma = 0.9$
four wear levels	three calls	how the world reacts	how much the future counts

$$\Big(\underbrace{s'}_{\text{the state next week}},\ \underbrace{r}_{\text{the reward}}\Big) = T(s, a)$$

The fleet sheet

HEALTHY

RUN profit	+95
breakdown	2%

The fleet sheet

$$\text{HEALTHY} \xrightarrow{\text{wear slips}} \text{WORN}$$

	HEALTHY	WORN
RUN profit	+95	+72
breakdown	2%	8%

The fleet sheet

$$\text{HEALTHY} \xrightarrow{\text{wear slips}} \text{WORN} \longrightarrow \text{SHAKY}$$

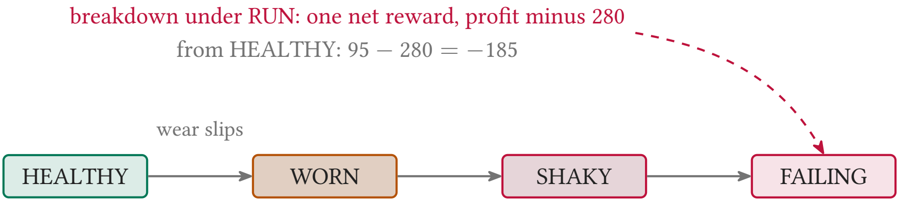
	HEALTHY	WORN	SHAKY
RUN profit	+95	+72	+40
breakdown	2%	8%	28%

The fleet sheet

$$\text{HEALTHY} \xrightarrow{\text{wear slips}} \text{WORN} \longrightarrow \text{SHAKY} \longrightarrow \text{FAILING}$$

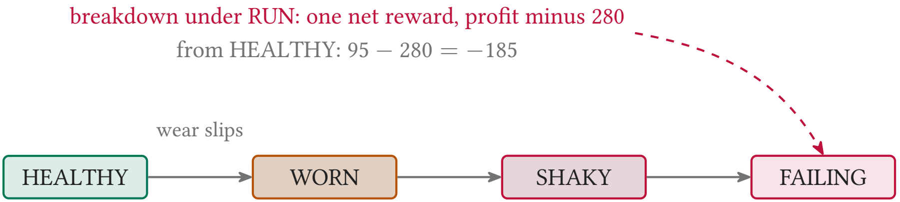
	HEALTHY	WORN	SHAKY	FAILING
RUN profit	+95	+72	+40	+16
breakdown	2%	8%	28%	55%

The fleet sheet



RUN profit	+95	+72	+40	+16
breakdown	2%	8%	28%	55%

The fleet sheet



RUN profit	+95	+72	+40	+16
breakdown	2%	8%	28%	55%

SERVICE: −50, a week in the shop (WORN back to HEALTHY 95% of the time)

The fleet sheet

breakdown under RUN: one net reward, profit minus 280

from HEALTHY: $95 - 280 = -185$

$\text{HEALTHY} \xrightarrow{\text{wear slips}} \text{WORN} \longrightarrow \text{SHAKY} \longrightarrow \text{FAILING}$

	HEALTHY	WORN	SHAKY	FAILING
RUN profit	$+95$	$+72$	$+40$	$+16$
breakdown	2%	8%	28%	55%

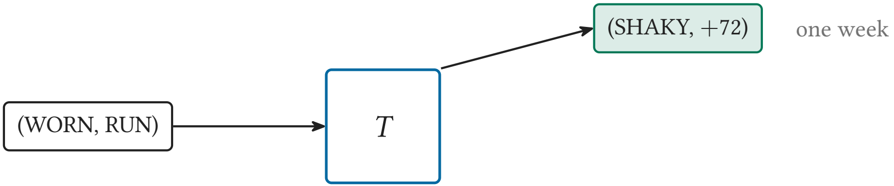
SERVICE: -50, a week in the shop (WORN back to HEALTHY 95% of the time)

REPLACE: -130, a brand-new HEALTHY van next week

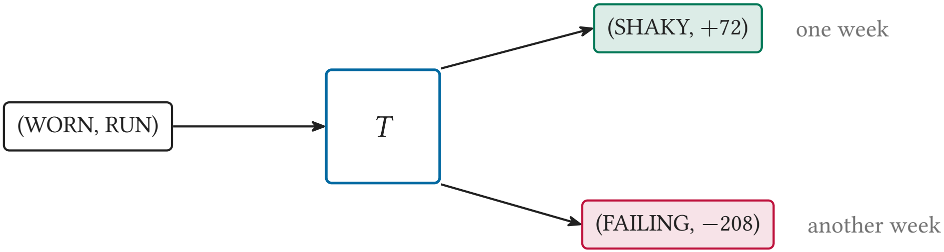
Same pair in, different week out

$$\boxed{(\text{WORN, RUN})} \longrightarrow \boxed{T}$$

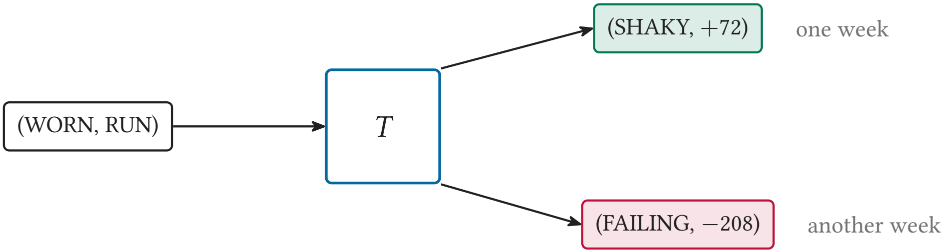
Same pair in, different week out



Same pair in, different week out



Same pair in, different week out



same input, different output: the dynamics T is random

Policies and trajectories

Playbook in, tape out

a policy (the playbook): $\pi : S \to A$, one call per state

Playbook in, tape out

a policy (the playbook): $\pi : S \to A$, one call per state

$\boxed{\text{HEALTHY} \to \text{RUN}}$ $\boxed{\text{WORN} \to \text{SERVICE}}$ $\boxed{\text{SHAKY} \to \text{REPLACE}}$ $\boxed{\text{FAILING} \to \text{REPLACE}}$

each week: $a_i = \pi(s_i)$, then $(s_{i+1}, r_i) = T(s_i, a_i)$

Playbook in, tape out

a policy (the playbook): $\pi : S \to A$, one call per state

HEALTHY $\to$ RUN	WORN $\to$ SERVICE	SHAKY $\to$ REPLACE	FAILING $\to$ REPLACE

each week: $a_i = \pi(s_i)$, then $(s_{i+1}, r_i) = T(s_i, a_i)$

wk 1
HEALTHY, RUN
$+95$

Playbook in, tape out

a policy (the playbook): $\pi : S \to A$, one call per state

HEALTHY → RUN | WORN → SERVICE | SHAKY → REPLACE | FAILING → REPLACE

each week: $a_i = \pi(s_i)$, then $(s_{i+1}, r_i) = T(s_i, a_i)$

wk 1	→	wk 2
HEALTHY, RUN		WORN, SERVICE
+95		−50

Playbook in, tape out

a policy (the playbook): $\pi : S \to A$, one call per state

$\boxed{\text{HEALTHY} \to \text{RUN}}$ $\boxed{\text{WORN} \to \text{SERVICE}}$ $\boxed{\text{SHAKY} \to \text{REPLACE}}$ $\boxed{\text{FAILING} \to \text{REPLACE}}$

each week: $a_i = \pi(s_i)$, then $(s_{i+1}, r_i) = T(s_i, a_i)$

$$\boxed{\begin{array}{c}\text{wk 1}\\ \text{HEALTHY, RUN}\\ +95\end{array}} \longrightarrow \boxed{\begin{array}{c}\text{wk 2}\\ \text{WORN, SERVICE}\\ -50\end{array}} \longrightarrow \boxed{\begin{array}{c}\text{wk 3}\\ \text{HEALTHY, RUN}\\ +95\end{array}}$$

Playbook in, tape out

a policy (the playbook): $\pi : S \to A$, one call per state

HEALTHY $\to$ RUN	WORN $\to$ SERVICE	SHAKY $\to$ REPLACE	FAILING $\to$ REPLACE

each week: $a_i = \pi(s_i)$, then $(s_{i+1}, r_i) = T(s_i, a_i)$

$$\boxed{\begin{matrix}\text{wk 1}\\ \text{HEALTHY, RUN}\\ +95\end{matrix}} \to \boxed{\begin{matrix}\text{wk 2}\\ \text{WORN, SERVICE}\\ -50\end{matrix}} \to \boxed{\begin{matrix}\text{wk 3}\\ \text{HEALTHY, RUN}\\ +95\end{matrix}} \to \boxed{\begin{matrix}\text{wk 4}\\ \text{HEALTHY, RUN}\\ +95\end{matrix}}$$

Playbook in, tape out

a policy (the playbook): $\pi : S \to A$, one call per state

HEALTHY → RUN | WORN → SERVICE | SHAKY → REPLACE | FAILING → REPLACE

each week: $a_i = \pi(s_i)$, then $(s_{i+1}, r_i) = T(s_i, a_i)$

wk 1 HEALTHY, RUN +95 → wk 2 WORN, SERVICE −50 → wk 3 HEALTHY, RUN +95 → wk 4 HEALTHY, RUN +95 → wk 5 WORN

the trajectory: $s_1, a_1, r_1,\ s_2, a_2, r_2, \ \dots$ one week per step, forever

Playbook in, tape out

a policy (the playbook): $\pi : S \to A$, one call per state

HEALTHY $\to$ RUN | WORN $\to$ SERVICE | SHAKY $\to$ REPLACE | FAILING $\to$ REPLACE

each week: $a_i = \pi(s_i)$, then $(s_{i+1}, r_i) = T(s_i, a_i)$

wk 1: HEALTHY, RUN, +95 $\to$ wk 2: WORN, SERVICE, −50 $\to$ wk 3: HEALTHY, RUN, +95 $\to$ wk 4: HEALTHY, RUN, +95 $\to$ wk 5: WORN

the trajectory: $s_1, a_1, r_1,\ s_2, a_2, r_2,\ \ldots$ one week per step, forever

the tape is random; another run, same start: (HEALTHY, RUN, −185) $\to$ FAILING

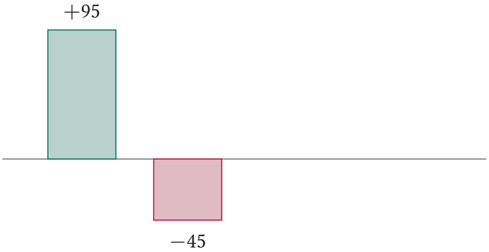
The goal: accumulating reward

The return: one number per tape

$$G_1 = r_1$$

The return: one number per tape

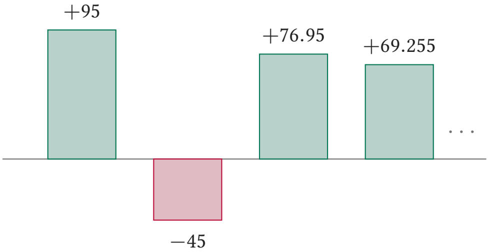
$$G_1 = \underbrace{r_1}_{\text{in full}} + \underbrace{\gamma\, r_2}_{\text{at } 90\%}$$



each week shrunk
by $\gamma = 0.9$

The return: one number per tape

$$G_1 = \underbrace{r_1}_{\text{in full}} + \underbrace{\gamma\, r_2}_{\text{at }90\%} + \underbrace{\gamma^2 r_3}_{\text{at }81\%} + \cdots$$

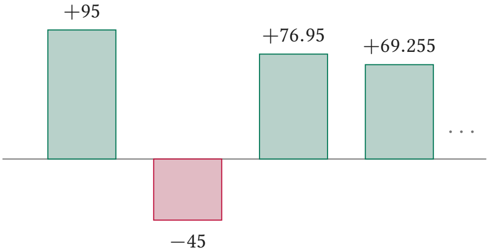


each week shrunk
by $\gamma = 0.9$

The return: one number per tape

$$G_1 = \underbrace{r_1}_{\text{in full}} + \underbrace{\gamma\, r_2}_{\text{at }90\%} + \underbrace{\gamma^2 r_3}_{\text{at }81\%} + \cdots$$

$$G_1 = 95 + 0.9\,(-50) + 0.81 \cdot 95 + 0.729 \cdot 95 + \cdots = 95 - 45 + 76.95 + 69.255 + \cdots$$

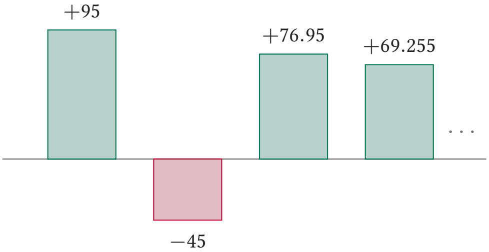


each week shrunk
by $\gamma = 0.9$

The return: one number per tape

$$G_1 = \underbrace{r_1}_{\text{in full}} + \underbrace{\gamma\, r_2}_{\text{at } 90\%} + \underbrace{\gamma^2 r_3}_{\text{at } 81\%} + \cdots$$

$$G_1 = 95 + 0.9\,(-50) + 0.81 \cdot 95 + 0.729 \cdot 95 + \cdots = 95 - 45 + 76.95 + 69.255 + \cdots$$



each week shrunk

by $\gamma = 0.9$

the shrinking factors keep

the never-ending sum finite

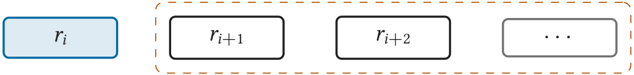
The recurrence

$$G_i = r_i + \gamma\, r_{i+1} + \gamma^2 r_{i+2} + \cdots$$

$$\boxed{r_i} \quad \boxed{r_{i+1}} \quad \boxed{r_{i+2}} \quad \boxed{\cdots}$$

The recurrence

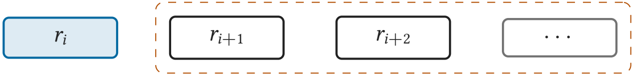
$$G_i = r_i + \gamma\left(r_{i+1} + \gamma\, r_{i+2} + \cdots \right)$$



G_{i+1}, discounted once by γ

The recurrence

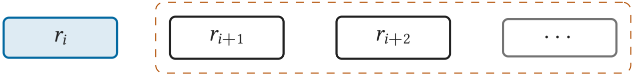
$$G_i = \underbrace{r_i}_{\text{this week}} + \gamma \underbrace{G_{i+1}}_{\text{everything after}}$$



G_{i+1}, discounted once by γ

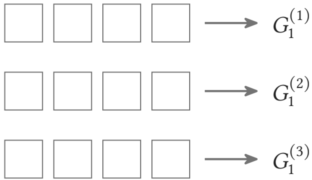
The recurrence

$$G_i = \underbrace{r_i}_{\text{this week}} + \gamma \underbrace{G_{i+1}}_{\text{everything after}}$$



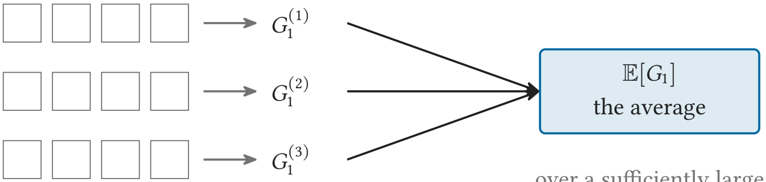
G_{i+1}, discounted once by γ

this week's reward $+ \gamma \times$ all the rest



same playbook, same start: a different G_1 every run

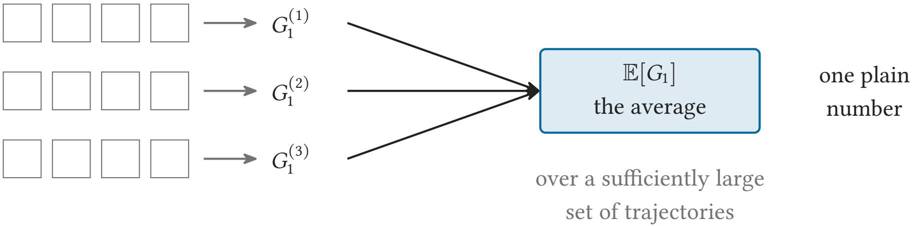
Maximize the average



same playbook, same start: a different G_1 every run

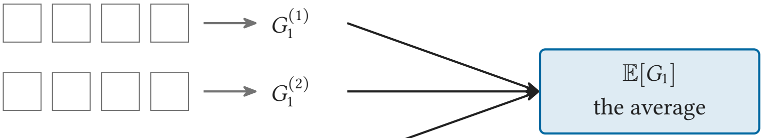
over a sufficiently large set of trajectories

Maximize the average



same playbook, same start: a different G_1 every run

Maximize the average



one plain number

over a sufficiently large set of trajectories

same playbook, same start: a different G_1 every run

the goal: find the policy π that maximizes $\mathbb{E}[G_1]$

The Q-function and the Bellman equation

The scorecard $Q^\star$

$$Q^\star(s, a) = \mathbb{E}\left[\, G_i \;\middle|\; s_i = s,\ a_i = a \,\right]$$

The scorecard $Q^\star$

$$Q^\star(s, a) = \mathbb{E}\left[\, G_i \;\middle|\; \underbrace{s_i = s,\ a_i = a}_{\text{in } s\text{, play } a\text{, optimal afterwards}} \right]$$

The scorecard $Q^\star$

$$Q^\star(s,a) = \mathbb{E}\Big[\, G_i \;\Big|\; \underbrace{s_i = s,\;\; a_i = a}_{\text{in } s\text{, play } a\text{, optimal afterwards}} \Big]$$

	RUN	SERVICE	REPLACE
HEALTHY	311.0	229.9	149.9

The scorecard $Q^\star$

$$Q^\star(s, a) = \mathbb{E}\left[\, G_i \;\middle|\; \underbrace{s_i = s,\ a_i = a}_{\text{in } s\text{, play } a\text{, optimal afterwards}} \right]$$

	RUN	SERVICE	REPLACE
HEALTHY	311.0	229.9	149.9
WORN	203.4	226.1	149.9

The scorecard $Q^\star$

$$Q^\star(s, a) = \mathbb{E}\left[\, G_i \;\middle|\; \underbrace{s_i = s,\;\; a_i = a}_{\text{in } s \text{, play } a \text{, optimal afterwards}} \right]$$

	RUN	SERVICE	REPLACE
HEALTHY	311.0	229.9	149.9
WORN	203.4	226.1	149.9
SHAKY	96.5	126.4	149.9

The scorecard $Q^\star$

$$Q^\star(s, a) = \mathbb{E}\Big[\, G_i \;\Big|\; \underbrace{s_i = s,\;\; a_i = a}_{\text{in } s\text{, play } a\text{, optimal afterwards}} \Big]$$

	RUN	SERVICE	REPLACE
HEALTHY	311.0	229.9	149.9
WORN	203.4	226.1	149.9
SHAKY	96.5	126.4	149.9
FAILING	−3.1	86.9	149.9

The scorecard $Q^\star$

$$Q^\star(s,a) = \mathbb{E}\big[\, G_i \;\big|\; \underbrace{s_i = s,\;\; a_i = a}_{\text{in } s\text{, play } a\text{, optimal afterwards}} \big]$$

$\pi^\star$: read each row, keep the call with the largest value

	RUN	SERVICE	REPLACE
HEALTHY	**311.0**	229.9	149.9
WORN	203.4	**226.1**	149.9
SHAKY	96.5	126.4	**149.9**
FAILING	−3.1	86.9	**149.9**

row by row: the kept call

The scorecard $Q^\star$

$$Q^\star(s, a) = \mathbb{E}\big[\, G_i \;\big|\; \underbrace{s_i = s,\;\; a_i = a}_{\text{in } s\text{, play } a\text{, optimal afterwards}} \big]$$

$\pi^\star$: read each row, keep the call with the largest value

	RUN	SERVICE	REPLACE
HEALTHY	311.0	229.9	149.9
WORN	203.4	226.1	149.9
SHAKY	96.5	126.4	149.9
FAILING	−3.1	86.9	149.9

row by row: the kept call

149.9 beats 126.4 by 23.5

The scorecard $Q^\star$

$$Q^\star(s,a) = \mathbb{E}\left[\, G_i \;\middle|\; \underbrace{s_i = s,\;\; a_i = a}_{\text{in } s\text{, play } a\text{, optimal afterwards}} \right]$$

$\pi^\star$: read each row, keep the call with the largest value

	RUN	SERVICE	REPLACE
HEALTHY	311.0	229.9	149.9
WORN	203.4	226.1	149.9
SHAKY	96.5	126.4	149.9
FAILING	−3.1	86.9	149.9

row by row: the kept call

149.9 beats 126.4 by 23.5

the optimal playbook replaces a van that still starts every morning

The Bellman equation

$$G_i = r_i + \gamma\, G_{i+1} \qquad \text{(every single week)}$$

$$(s_i,\ a_i) \xrightarrow{\text{pays } r_i} (s_{i+1},\ a_{i+1}) \qquad \text{counted at } \gamma$$

The Bellman equation

$$Q^\star(s_i, a_i) = \mathbb{E}\left[\, r_i + \gamma\, G_{i+1} \,\right]$$ (average both sides)

$$(s_i,\ a_i) \xrightarrow{\text{pays } r_i} (s_{i+1},\ a_{i+1})$$ counted at γ

The Bellman equation

$$\boxed{Q^\star(s_i, a_i) = \mathbb{E}\Big[\underbrace{r_i}_{\text{this week's cash}} + \gamma \underbrace{Q^\star(s_{i+1}, a_{i+1})}_{\text{next week's value}}\Big]}$$

replacing G_{i+1} by its average $Q^\star(s_{i+1}, a_{i+1})$: averaging in two rounds

$$(s_i,\ a_i) \xrightarrow{\text{pays } r_i} (s_{i+1},\ a_{i+1}) \qquad \text{counted at } \gamma$$

The Bellman equation

$$\boxed{Q^\star(s_i, a_i) = \mathbb{E}\Big[\; \underbrace{r_i}_{\text{this week's cash}} + \gamma\, \underbrace{Q^\star(s_{i+1}, a_{i+1})}_{\text{next week's value}} \Big]}$$

replacing G_{i+1} by its average $Q^\star(s_{i+1}, a_{i+1})$: averaging in two rounds

$$(s_i,\ a_i) \xrightarrow{\text{pays } r_i} (s_{i+1},\ a_{i+1}) \qquad \text{counted at } \gamma$$

one equation per cell: twelve equations pin down the table

SARSA: learning from the logbook

Three bookkeepers, one bad week

nobody prints the odds for your van: fill the (WORN, RUN) card from lived weeks

(WORN, RUN) pays	+72
Overwriter card = last week	72
Filing cabinet true average of every week	72
Nudger move 15% toward the week	72

Three bookkeepers, one bad week

nobody prints the odds for your van: fill the (WORN, RUN) card from lived weeks

(WORN, RUN) pays	+72	+72
Overwriter card = last week	72	72
Filing cabinet true average of every week	72	72
Nudger move 15% toward the week	72	72

Three bookkeepers, one bad week

nobody prints the odds for your van: fill the (WORN, RUN) card from lived weeks

(WORN, RUN) pays	+72	+72	+72
Overwriter card = last week	72	72	72
Filing cabinet true average of every week	72	72	72
Nudger move 15% toward the week	72	72	72

Three bookkeepers, one bad week

nobody prints the odds for your van: fill the (WORN, RUN) card from lived weeks

(WORN, RUN) pays	+72	+72	+72	−208
Overwriter card = last week	72	72	72	−208
Filing cabinet true average of every week	72	72	72	2
Nudger move 15% toward the week	72	72	72	30

Three bookkeepers, one bad week

nobody prints the odds for your van: fill the (WORN, RUN) card from lived weeks

(WORN, RUN) pays	+72	+72	+72	−208	+72
Overwriter card = last week	72	72	72	−208	72
Filing cabinet true average of every week	72	72	72	2	16
Nudger move 15% toward the week	72	72	72	30	36.3

Three bookkeepers, one bad week

nobody prints the odds for your van: fill the (WORN, RUN) card from lived weeks

(WORN, RUN) pays	+72	+72	+72	−208	+72	
Overwriter card = last week	72	72	72	−208	72	hysterical
Filing cabinet true average of every week	72	72	72	2	16	right, but stores every card
Nudger move 15% toward the week	72	72	72	30	36.3	calm, and no cabinet

Three bookkeepers, one bad week

nobody prints the odds for your van: fill the (WORN, RUN) card from lived weeks

(WORN, RUN) pays	+72	+72	+72	−208	+72	
Overwriter card = last week	72	72	72	−208	72	hysterical
Filing cabinet true average of every week	72	72	72	2	16	right, but stores every card
Nudger move 15% toward the week	72	72	72	30	36.3	calm, and no cabinet

$$q(s_i, a_i) \leftarrow q(s_i, a_i) + \alpha \left[r_i - q(s_i, a_i) \right]$$

(the Nudger, written down)

Three bookkeepers, one bad week

nobody prints the odds for your van: fill the (WORN, RUN) card from lived weeks

(WORN, RUN) pays	+72	+72	+72	−208	+72	
Overwriter card = last week	72	72	72	−208	72	hysterical
Filing cabinet true average of every week	72	72	72	2	16	right, but stores every card
Nudger move 15% toward the week	72	72	72	30	36.3	calm, and no cabinet

$$q(s_i, a_i) \leftarrow \underbrace{q(s_i, a_i)}_{\text{the card}} + \underbrace{\alpha}_{0.15} \underbrace{\left[r_i - q(s_i, a_i) \right]}_{\text{the surprise}}$$

Three bookkeepers, one bad week

nobody prints the odds for your van: fill the (WORN, RUN) card from lived weeks

(WORN, RUN) pays	+72	+72	+72	−208	+72	
Overwriter card = last week	72	72	72	−208	72	hysterical
Filing cabinet true average of every week	72	72	72	2	16	right, but stores every card
Nudger move 15% toward the week	72	72	72	30	36.3	calm, and no cabinet

$$q(s_i, a_i) \leftarrow \underbrace{q(s_i, a_i)}_{\text{the card}} + \underbrace{\alpha}_{0.15} \underbrace{\left[r_i - q(s_i, a_i) \right]}_{\text{the surprise}}$$

nudge, never overwrite: one bad week must not erase a year

The alpha knob

same five weeks, three step sizes

The alpha knob

same five weeks, three step sizes

$\alpha = 0.9$

72

the -208 week lands

-180

then $+72$ lands

46.8

whiplash

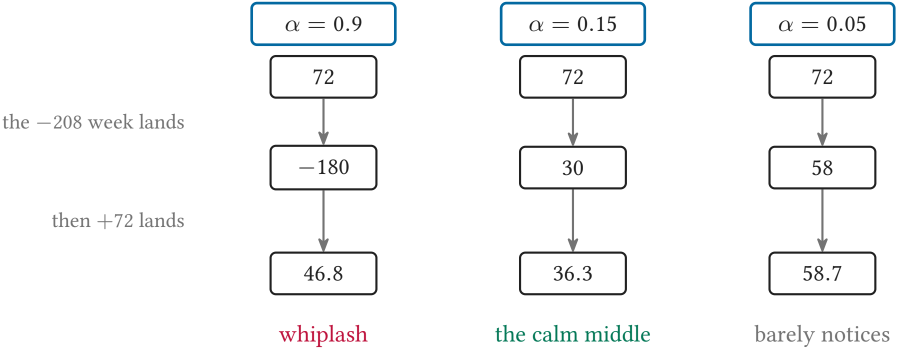
The alpha knob

same five weeks, three step sizes

	$\alpha = 0.9$	$\alpha = 0.05$
	72	72
the -208 week lands	$\downarrow$	$\downarrow$
	-180	58
then $+72$ lands	$\downarrow$	$\downarrow$
	46.8	58.7
	whiplash	barely notices

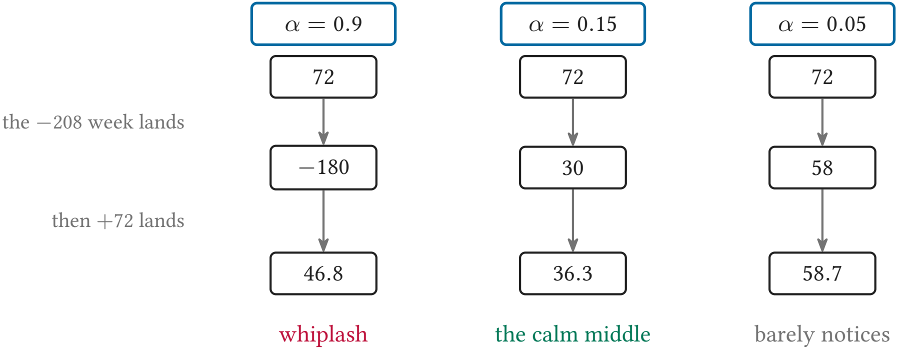
The alpha knob

same five weeks, three step sizes



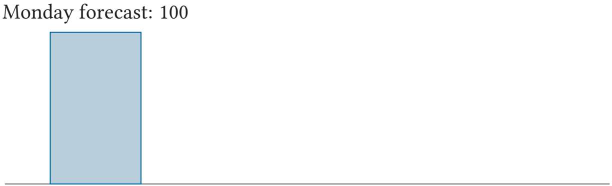
The alpha knob

same five weeks, three step sizes

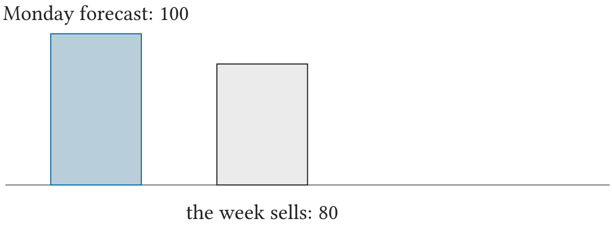


small steady steps, never overwrite

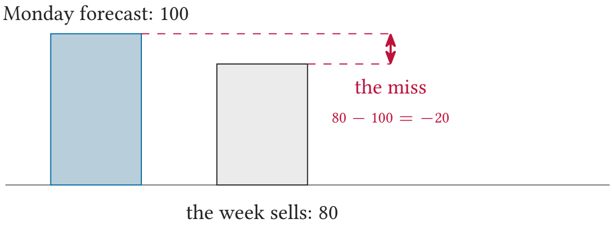
The planner’s Monday



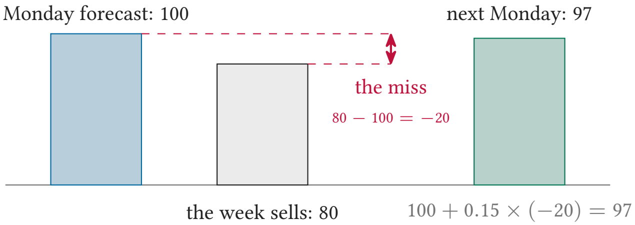
The planner's Monday



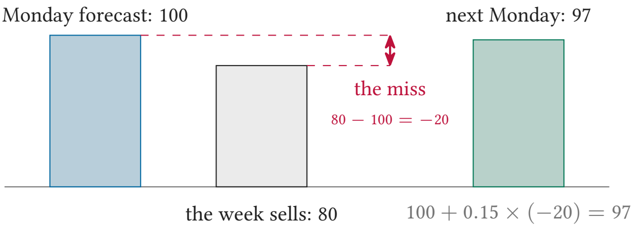
The planner's Monday



The planner's Monday

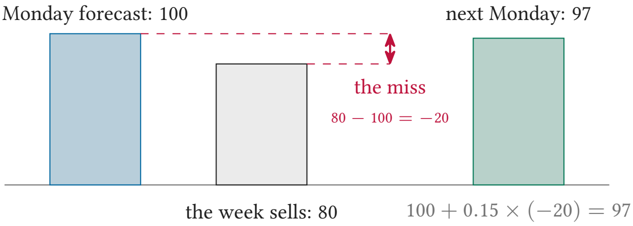


The planner's Monday



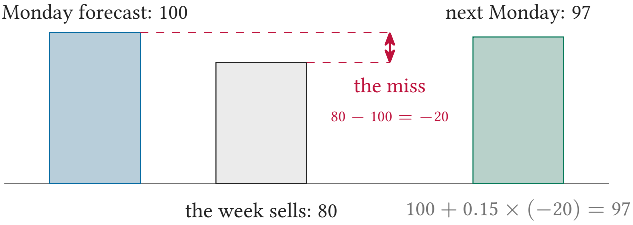
$$\text{new forecast} = \text{old forecast} + 0.15 \times \text{miss}$$

The planner's Monday



$$\text{new forecast} = \underbrace{\text{old forecast}}_{q(s_i,\, a_i)} + \underbrace{0.15}_{\alpha} \times \underbrace{\text{miss}}_{r_i - q(s_i, a_i)}$$

The planner's Monday

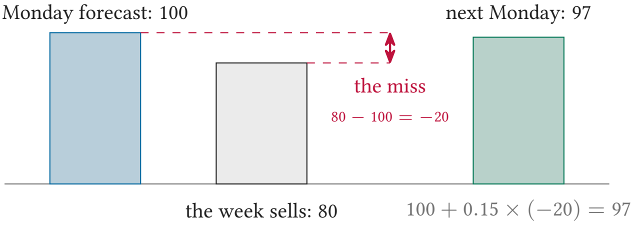


the planner's name:

exponential smoothing

$$\text{new forecast} = \underbrace{\text{old forecast}}_{q(s_i,\, a_i)} + \underbrace{0.15}_{\alpha} \times \underbrace{\text{miss}}_{r_i - q(s_i, a_i)}$$

The planner's Monday



the planner's name:

exponential smoothing

$$\text{new forecast} = \underbrace{\text{old forecast}}_{q(s_i,\, a_i)} + \underbrace{0.15}_{\alpha} \times \underbrace{\text{miss}}_{r_i - q(s_i, a_i)}$$

SARSA forecasts the value of a call the way a planner forecasts sales

The cash-only ledger lies

the cash-only ledger

REPLACE: −130, every single week

worst call on the sheet

The cash-only ledger lies

the cash-only ledger

REPLACE: −130, every single week

worst call on the sheet

SERVICE: −50

The cash-only ledger lies

the cash-only ledger

REPLACE: −130, every single week

worst call on the sheet

SERVICE: −50

the scorecard $Q^\star$ you saw

REPLACE crowned at SHAKY and FAILING: 149.9

The cash-only ledger lies

the cash-only ledger

REPLACE: −130, every single week

worst call on the sheet

SERVICE: −50

the scorecard $Q^{\star}$ you saw

REPLACE crowned at SHAKY and FAILING: 149.9

SERVICE crowned at WORN: 226.1

The cash-only ledger lies

the cash-only ledger

REPLACE: −130, every single week

worst call on the sheet

SERVICE: −50

the scorecard $Q^\star$ you saw

REPLACE crowned at SHAKY and FAILING: 149.9

SERVICE crowned at WORN: 226.1

what does the cash-only ledger miss?

The cash-only ledger lies

the cash-only ledger

REPLACE: −130, every single week

worst call on the sheet

SERVICE: −50

the scorecard $Q^\star$ you saw

REPLACE crowned at SHAKY and FAILING: 149.9

SERVICE crowned at WORN: 226.1

what does the cash-only ledger miss?

one week $\xrightarrow{\text{the same cash}}$ a HEALTHY van left in the lot

The cash-only ledger lies

the cash-only ledger

REPLACE: −130, every single week

worst call on the sheet

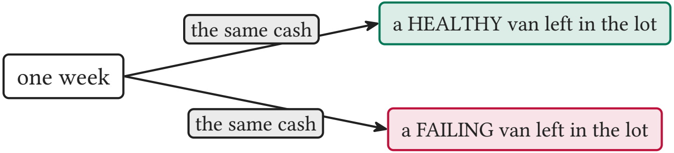
SERVICE: −50

the scorecard $Q^\star$ you saw

REPLACE crowned at SHAKY and FAILING: 149.9

SERVICE crowned at WORN: 226.1

what does the cash-only ledger miss?



The cash-only ledger lies

the cash-only ledger

REPLACE: −130, every single week

worst call on the sheet

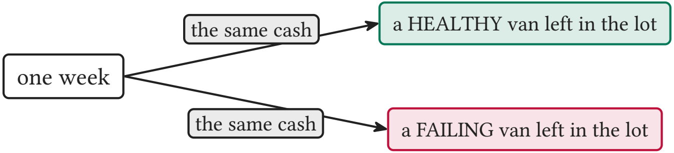
SERVICE: −50

the scorecard $Q^\star$ you saw

REPLACE crowned at SHAKY and FAILING: 149.9

SERVICE crowned at WORN: 226.1

what does the cash-only ledger miss?



same cash, different futures: so what is the leftover van worth?

Grade the week like a trade-in

S	**A**	**R**	**S**	**A**	
s_i	a_i	r_i	s_{i+1}	a_{i+1}	one logbook line

Grade the week like a trade-in

S	A	R	S	A	
s_i	a_i	r_i	s_{i+1}	a_{i+1}	one logbook line
		cash	the pair you land on		

trade-in price: $\gamma \times q(s_{i+1}, a_{i+1})$

Grade the week like a trade-in

S	A	R	S	A	
s_i	a_i	r_i	s_{i+1}	a_{i+1}	one logbook line
		cash	the pair you land on		

trade-in price: $\gamma \times q(s_{i+1}, a_{i+1})$

$$\text{the week's worth} = \underbrace{r_i}_{\text{cash}} + \underbrace{\gamma\, q\big(s_{i+1}, a_{i+1}\big)}_{\text{trade-in, at your own price}}$$

Grade the week like a trade-in

S	A	R	S	A	
s_i	a_i	r_i	s_{i+1}	a_{i+1}	one logbook line
		cash	the pair you land on		

trade-in price: $\gamma \times q(s_{i+1}, a_{i+1})$

$$\text{the better target} = \underbrace{r_i + \gamma\, q\big(s_{i+1}, a_{i+1}\big)}_{\text{cash, plus the van you keep}}$$

the surprise was your guess against cash: now aim it at this

Grade the week like a trade-in

S	A	R	S	A	
s_i	a_i	r_i	s_{i+1}	a_{i+1}	one logbook line
		cash	the pair you land on		

$\boxed{\text{trade-in price: } \gamma \times q(s_{i+1}, a_{i+1})}$

$$\text{the better target} = \underbrace{r_i + \gamma\, q\big(s_{i+1}, a_{i+1}\big)}_{\text{cash, plus the van you keep}}$$

the surprise was your guess against cash: now aim it at this

upgrade the target: from cash alone to cash + trade-in

Price it from your own book

wait: you price the leftover van with q, the very book you are still learning

Price it from your own book

wait: you price the leftover van with q, the very book you are still learning

Price it from your own book

wait: you price the leftover van with q, the very book you are still learning

a dealer's price book
this month, still being revised

→

prices today's trade-in
from that book, anyway

nobody refuses to price a car because the book is not final

Price it from your own book

wait: you price the leftover van with q, the very book you are still learning

a dealer's price book
this month, still being revised

prices today's trade-in
from that book, anyway

nobody refuses to price a car because the book is not final

week 0

0

each nudge: less wrong

many weeks

$\approx Q^{\star}$

Price it from your own book

wait: you price the leftover van with q, the very book you are still learning

a dealer's price book
this month, still being revised

prices today's trade-in
from that book, anyway

nobody refuses to price a car because the book is not final

week 0

0

each nudge: less wrong

many weeks

$\approx Q^{\star}$

the book is wrong today, and that is fine: every nudge makes it less wrong

Write the line yourself

cash
r_i

value where you land
$\gamma\, q(s_{i+1}, a_{i+1})$

what you believed
$q(s_i, a_i)$

Write the line yourself

cash
r_i

value where you land
$\gamma\, q(s_{i+1}, a_{i+1})$

what you believed
$q(s_i, a_i)$

$$\text{new guess} = \text{old guess} + \alpha\Big[\,\rule{2em}{0.4pt} + \gamma\,\rule{6em}{0.4pt} - \rule{4em}{0.4pt}\,\Big]$$

Write the line yourself

cash	value where you land	what you believed
r_i	$\gamma\, q(s_{i+1}, a_{i+1})$	$q(s_i, a_i)$

$$q(s_i, a_i) \leftarrow q(s_i, a_i) + \alpha \left[\, r_i + \gamma\, q(s_{i+1}, a_{i+1}) - q(s_i, a_i) \,\right]$$

Write the line yourself

cash	value where you land	what you believed
r_i	$\gamma\, q(s_{i+1}, a_{i+1})$	$q(s_i, a_i)$

$$q(s_i, a_i) \leftarrow q(s_i, a_i) + \alpha \underbrace{\left[\, r_i + \gamma\, q(s_{i+1}, a_{i+1}) - q(s_i, a_i) \,\right]}_{\text{the surprise: upgraded target minus your guess}}$$

Write the line yourself

cash
r_i

value where you land
$\gamma\, q(s_{i+1}, a_{i+1})$

what you believed
$q(s_i, a_i)$

$$q(s_i, a_i) \leftarrow q(s_i, a_i) + \alpha \underbrace{\left[\, r_i + \gamma\, q(s_{i+1}, a_{i+1}) - q(s_i, a_i) \,\right]}_{\text{the surprise: upgraded target minus your guess}}$$

$\alpha = 0.15$, constant · $\varepsilon = 0.15$: a random call in 15% of weeks · a_{i+1}: the call you actually make, never a max

Write the line yourself

cash
r_i

value where you land
$\gamma\, q(s_{i+1}, a_{i+1})$

what you believed
$q(s_i, a_i)$

$$q(s_i, a_i) \leftarrow q(s_i, a_i) + \alpha \underbrace{\left[\, r_i + \gamma\, q(s_{i+1}, a_{i+1}) - q(s_i, a_i) \,\right]}_{\text{the surprise: upgraded target minus your guess}}$$

$\alpha = 0.15$, constant · $\varepsilon = 0.15$: a random call in 15% of weeks · a_{i+1}: the call you actually make, never a max

you did not memorize SARSA: you wrote it

Bellman becomes SARSA

$$Q^{\star}(s_i, a_i) = \mathbb{E}\left[\, r_i + \gamma\, Q^{\star}(s_{i+1}, a_{i+1}) \,\right]$$

start from Bellman: an average over all the weeks that could happen

Bellman becomes SARSA

$$Q^\star(s_i, a_i) \approx \underbrace{r_i + \gamma\, Q^\star(s_{i+1}, a_{i+1})}_{\text{the one week that arrived, not the average}}$$

erase the average: keep the single week that actually arrived

Bellman becomes SARSA

$$q(s_i, a_i) \approx r_i + \gamma\, q(s_{i+1}, a_{i+1}) \qquad \text{(your book, not } Q^\star\text{)}$$

erase the unknown truth $Q^\star$: look the value up in your own book q

Bellman becomes SARSA

$$\boxed{q(s_i, a_i) \leftarrow q(s_i, a_i) + \alpha \big[\, r_i + \gamma\, q(s_{i+1}, a_{i+1}) - q(s_i, a_i) \,\big]}$$

wrap it in the nudge: step toward the target, never jump to it

Bellman becomes SARSA

$$\boxed{q(s_i, a_i) \leftarrow q(s_i, a_i) + \alpha \left[r_i + \gamma\, q(s_{i+1}, a_{i+1}) - q(s_i, a_i) \right]}$$

wrap it in the nudge: step toward the target, never jump to it

Bellman, with the expectation replaced by an arriving average

One week, one nudge

twelve guesses, all start at 0

	RUN	SERVICE	REPLACE
HEALTHY	0	0	0
WORN	0	0	0
SHAKY	0	0	0
FAILING	0	0	0

One week, one nudge

week one in the logbook: (HEALTHY, RUN, +95, WORN, SERVICE)

twelve guesses, all start at 0

	RUN	SERVICE	REPLACE
HEALTHY	0	0	0
WORN	0	0	0
SHAKY	0	0	0
FAILING	0	0	0

One week, one nudge

week one in the logbook: (HEALTHY, RUN, +95, WORN, SERVICE)

$$\text{evidence} = r_i + \gamma\, q(\text{WORN}, \text{SERVICE}) = 95 + 0.9 \times 0 = 95$$

twelve guesses, all start at 0

	RUN	SERVICE	REPLACE
HEALTHY	0	0	0
WORN	0	0	0
SHAKY	0	0	0
FAILING	0	0	0

One week, one nudge

week one in the logbook: (HEALTHY, RUN, +95, WORN, SERVICE)

$$\text{evidence} = r_i + \gamma\, q(\text{WORN}, \text{SERVICE}) = 95 + 0.9 \times 0 = 95$$

$$\text{surprise} = 95 - 0 = 95$$

nudge: $+\,0.15 \times 95$

twelve guesses, all start at 0

	RUN	SERVICE	REPLACE
HEALTHY	0	0	0
WORN	0	0	0
SHAKY	0	0	0
FAILING	0	0	0

One week, one nudge

week one in the logbook: (HEALTHY, RUN, +95, WORN, SERVICE)

$$\text{evidence} = r_i + \gamma\, q(\text{WORN}, \text{SERVICE}) = 95 + 0.9 \times 0 = 95$$

$$\text{surprise} = 95 - 0 = 95$$

nudge: $+\,0.15 \times 95$

twelve guesses, all start at 0

one cell moves, eleven stay put

	RUN	SERVICE	REPLACE
HEALTHY	14.25	0	0
WORN	0	0	0
SHAKY	0	0	0
FAILING	0	0	0

One week, one nudge

week one in the logbook: (HEALTHY, RUN, +95, WORN, SERVICE)

$$\text{evidence} = r_i + \gamma\, q(\text{WORN}, \text{SERVICE}) = 95 + 0.9 \times 0 = 95$$

$$\text{surprise} = 95 - 0 = 95$$

nudge: $+\,0.15 \times 95$

twelve guesses, all start at 0

one cell moves, eleven stay put

a few hundred weeks later:

the three bands emerge

	RUN	SERVICE	REPLACE
HEALTHY	14.25	0	0
WORN	0	0	0
SHAKY	0	0	0
FAILING	0	0	0

One week, one nudge

week one in the logbook: (HEALTHY, RUN, +95, WORN, SERVICE)

$$\text{evidence} = r_i + \gamma\, q(\text{WORN}, \text{SERVICE}) = 95 + 0.9 \times 0 = 95$$

$$\text{surprise} = 95 - 0 = 95$$

nudge: $+\,0.15 \times 95$

twelve guesses, all start at 0

one cell moves, eleven stay put

a few hundred weeks later:

the three bands emerge

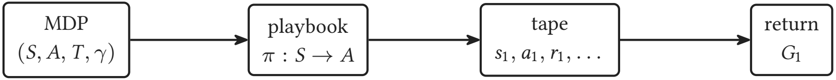
	RUN	SERVICE	REPLACE
HEALTHY	14.25	0	0
WORN	0	0	0
SHAKY	0	0	0
FAILING	0	0	0

the cells settle below $Q^\star$ and keep jittering, but each row keeps the right call

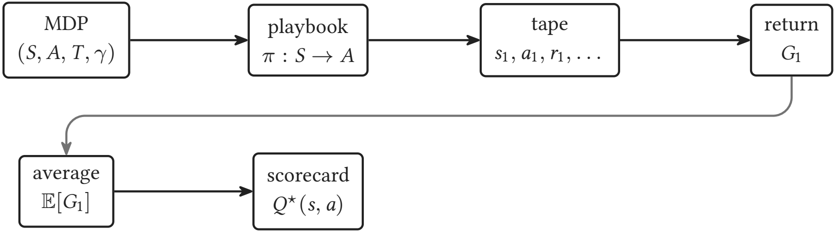
The whole story

$$
\begin{tikzcd}
\boxed{\begin{array}{c}\text{MDP}\\(S, A, T, \gamma)\end{array}} \arrow[r] & \boxed{\begin{array}{c}\text{playbook}\\ \pi : S \to A\end{array}}
\end{tikzcd}
$$

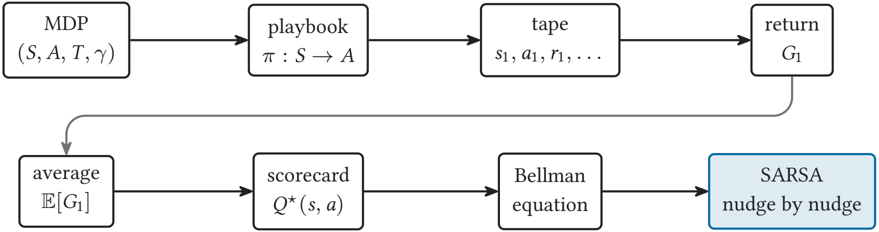
The whole story



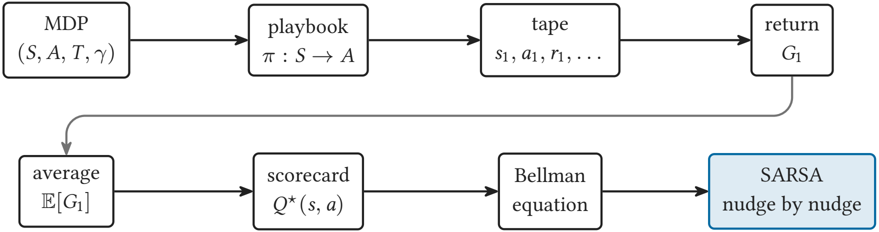
The whole story



The whole story

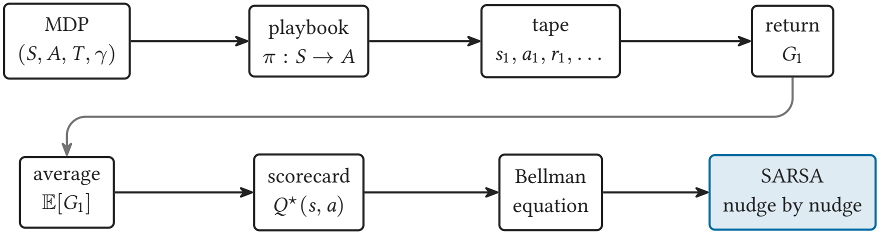


The whole story



SARSA solves the Bellman equation by averaging evidence instead of computing expectations

The whole story



SARSA solves the Bellman equation by averaging evidence instead of computing expectations

play it: `https://sml-fs26.github.io/classic_rl/repair-or-replace/`

Appendix: why the nudge works

The Robbins-Monro iteration

find the number $x^\star$ that equals the average of noisy samples, one at a time

The Robbins-Monro iteration

find the number $x^\star$ that equals the average of noisy samples, one at a time

$$x \leftarrow x + \alpha_n \,(\text{sample} - x)$$

The Robbins-Monro iteration

find the number $x^\star$ that equals the average of noisy samples, one at a time

$$x \leftarrow x + \underbrace{\alpha_n}_{\text{step } n} \underbrace{(\text{sample} - x)}_{\text{the miss}}$$

The Robbins-Monro iteration

find the number $x^\star$ that equals the average of noisy samples, one at a time

$$x \leftarrow x + \underbrace{\alpha_n}_{\text{step } n} \underbrace{(\text{sample} - x)}_{\text{the miss}}$$

$\alpha_1 + \alpha_2 + \alpha_3 + \cdots$ unbounded (can travel from any start)

The Robbins-Monro iteration

find the number $x^\star$ that equals the average of noisy samples, one at a time

$$x \leftarrow x + \underbrace{\alpha_n}_{\text{step } n} \underbrace{(\text{sample} - x)}_{\text{the miss}}$$

$\alpha_1 + \alpha_2 + \alpha_3 + \cdots$ unbounded (can travel from any start)

$\alpha_1^2 + \alpha_2^2 + \alpha_3^2 + \cdots$ finite (the noise eventually cancels)

The Robbins-Monro iteration

find the number $x^\star$ that equals the average of noisy samples, one at a time

$$x \leftarrow x + \underbrace{\alpha_n}_{\text{step } n} \underbrace{(\text{sample} - x)}_{\text{the miss}}$$

$\alpha_1 + \alpha_2 + \alpha_3 + \cdots$ unbounded (can travel from any start)

$\alpha_1^2 + \alpha_2^2 + \alpha_3^2 + \cdots$ finite (the noise eventually cancels)

$\alpha_n = 1/n$ satisfies both demands

The Robbins-Monro iteration

find the number $x^\star$ that equals the average of noisy samples, one at a time

$$x \leftarrow x + \underbrace{\alpha_n}_{\text{step } n} \underbrace{(\text{sample} - x)}_{\text{the miss}}$$

$\alpha_1 + \alpha_2 + \alpha_3 + \cdots$ unbounded (can travel from any start)

$\alpha_1^2 + \alpha_2^2 + \alpha_3^2 + \cdots$ finite (the noise eventually cancels)

$\alpha_n = 1/n$ satisfies both demands

SARSA is Robbins-Monro on the Bellman equation, cell by cell

The running average

A_n: the average of the first n observations $x_1, \ldots, x_n$

The running average

A_n: the average of the first n observations $x_1, \ldots, x_n$

$$A_n = A_{n-1} + \tfrac{1}{n}\left(x_n - A_{n-1}\right)$$

The running average

A_n: the average of the first n observations $x_1, \ldots, x_n$

$$A_n = \underbrace{A_{n-1}}_{\text{average so far}} + \frac{1}{n} \underbrace{\left(x_n - A_{n-1}\right)}_{\text{one share of the surprise}}$$

The running average

A_n: the average of the first n observations $x_1, \ldots, x_n$

$$A_n = \underbrace{A_{n-1}}_{\text{average so far}} + \frac{1}{n} \underbrace{\left(x_n - A_{n-1}\right)}_{\text{one share of the surprise}}$$

hold α constant instead of $1/n$: each old week fades by $(1 - \alpha)$

The running average

A_n: the average of the first n observations $x_1, \ldots, x_n$

$$A_n = \underbrace{A_{n-1}}_{\text{average so far}} + \frac{1}{n} \underbrace{\big(x_n - A_{n-1}\big)}_{\text{one share of the surprise}}$$

hold α constant instead of $1/n$: each old week fades by $(1-\alpha)$

SARSA is a running average of the weekly evidence